

Tiny URL Shortner

Requirements

Non-Functional

- Scalability
- Availability
- Performance

Functional

- Shorten the URL
- Redirect to the Original URL
- Generate Unique Short Codes

Estimations

Assume :

- Time Period : 5 years
- Active users : 10M
- Read : Write : 1000:1
- Expected Traffic :

Read QPS : 10,000/sec
Write QPS : 10/sec

Final Estimations :

- Cache : 64 GB
- Data-Base Servers : 2 (read , write server)
- 2 Application Servers

DataBase

• Two Database Servers are required:

1. Database Server 1 (Read Database)

- Handles **GET requests** (URL lookups)
- Stores URL mappings and provides fast retrieval of original URLs using short_code
- Can be optimized with indexing on short_code

2. Database Server 2 (Write Database)

- Handles POST requests (creation of new short URLs).
- Stores newly generated short URL mappings.

- Base62 Encoding is used for generating short URLs

Short URL	Original URL
Varchar(10)	Text

URL Table Structure

API Endpoints

API Endpoint	Method	Purpose
/api/v1/shorten	POST	Generate a short URL from the original long URL
/api/v1/{short_code}	GET	Redirect user to the original URL using the short code

Architecture

